Name: Muhammad Irfan

AI Prompting Techniques & Model Comparison

1. ZERO SHOT PROMPTING

Prompt:
Summarize the following 500-word technical article on "The Basics of Blockchain Technology" for undergraduate students. Write the summary in simple and clear language, highlight the key concepts, and keep it concise and easy to understand. Avoid technical jargon where possible and focus only on the most important points from the article.

CHATGPT

Blockchain technology is a digital system used to store and share information securely. It works like a chain of blocks, where each block contains data and is connected to the previous one. This makes the information difficult to change or hack.

One of the main features of blockchain is decentralization. Instead of being controlled by a single person or company, the data is shared across many computers. This increases security and reduces the risk of failure. Blockchain is also transparent because all users in the network can view the recorded transactions. Once information is added, it cannot be easily changed, which makes the system trustworthy and reliable.

Cryptocurrency, such as Bitcoin, is one of the most common uses of blockchain technology. However, blockchain is also used in areas like banking, healthcare, supply chains, and voting systems to improve security and efficiency. Overall, blockchain technology provides a safe, transparent, and efficient way to store and transfer data without needing a central authority.

GEMINI

Blockchain technology is essentially a shared, digital ledger that records transactions across a network of computers. Unlike a traditional bank database managed by a single institution, blockchain is

decentralized, meaning no single person or entity has total control. Here are the key concepts that explain how it works:

- **1. The Structure: Blocks and Chains:** Information is stored in groups called "blocks." Each block contains a list of recent transactions. Once a block is full, it is "chained" to the previous block using a unique digital fingerprint called a hash. This creates a chronological timeline of data that cannot be easily altered.
- **2. Decentralization and Transparency:** Instead of one central server, every computer in the network (called a node) keeps a copy of the entire blockchain. This means if one computer fails or is hacked, the rest of the network remains secure and the data stays intact. Everyone on the network can see the history of transactions, ensuring transparency.
- **3. Security Through Consensus:** Before a new block is added to the chain, the network must agree that the transactions inside it are valid. This process is called consensus. Because each block is linked to the one before it with a unique hash, changing even one piece of data in an old block would change its hash and break the entire chain, making fraud very easy to detect.
- **4. Key Benefits:** Immutability (almost impossible to change), Efficiency (removes middlemen), and Trust (relies on math and code).

In summary, blockchain is a highly secure, transparent way of recording information that makes it almost impossible to cheat the system. While it is the foundation for cryptocurrencies like Bitcoin, its ability to provide a "single source of truth" makes it useful for everything from tracking supply chains to digital voting.

CLAUDE

Got it! Here's a concise, student-friendly summary of Blockchain Technology written from scratch:

What Is Blockchain? A blockchain is like a digital notebook that records information. But unlike a regular notebook, no single person owns or controls it. Instead, thousands of computers around the world share and maintain it together. Once something is written in this notebook, it cannot be erased or changed.

How Does It Work? Information is grouped into blocks. Each block holds a batch of records. When a block is full, it gets linked to the previous block, forming a chain. Every block contains three key things: Data, A hash (unique code), and The previous block's hash.

Why Is It Secure? Because each block is connected to the one before it, tampering with one block would break the entire chain. On top of that, the blockchain isn't stored in one place - it lives on thousands of computers simultaneously.

Who Keeps It Running? No single authority controls a blockchain. Instead, it runs on a peer-to-peer network. Before any new block is added, the majority of participants must agree (consensus).

Where Is It Used? Healthcare, Supply chains, Voting systems, and Contracts (smart contracts).

Why Does It Matter? Blockchain solves a fundamental problem: how do you trust information shared between strangers? Blockchain removes that need by making the record itself trustworthy through transparency and mathematics.

COPILOT

Blockchain Basics - Simplified

- **Definition:** Blockchain is a digital system that records information in a secure and transparent way. Think of it as a shared notebook that everyone can see but no one can erase or change without agreement.
- **Structure:** Information is stored in "blocks." Each block contains data and is linked to the previous one, forming a "chain."
- **Decentralization:** Unlike traditional systems controlled by banks or governments, blockchain has no single owner. Instead, it's managed by a network of computers working together.
- **Security:** Blockchain uses cryptography to keep data safe. Once information is added, it's nearly impossible to alter without everyone noticing.
- **Transparency:** Everyone in the network can see the records. This builds trust.
- **Applications:** Cryptocurrency, Smart Contracts, Supply Chains, Voting Systems.
- **Advantages:** Secure and trustworthy, Reduces need for middlemen, Increases efficiency.
- **Challenges:** Uses a lot of energy, Can be slow for large-scale use, Still evolving.

Key takeaway: Blockchain is a powerful way to store and share information securely without needing a central authority.

2. FEW SHOT PROMPTING

Prompt:
You are part of a content curation team for an educational platform that delivers quick summaries of research papers and technical articles to undergraduate students. Your task is to summarize technical content in a simple, concise, and student-friendly way.

Example 1 Input: Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, and decision-making. AI systems use algorithms and data to improve their performance over time. Output: Artificial Intelligence (AI) allows machines to imitate human thinking and decision-making. It uses data and algorithms to learn and improve performance over time. AI is commonly used in areas like automation, recommendations, and problem-solving.

Example 2 Input: Cloud computing is the delivery of computing services such as storage, servers,

and software over the internet. It helps organizations reduce costs, improve scalability, and access resources remotely.
Output: Cloud computing provides services like storage, servers, and software through the internet. It helps businesses save money, scale resources easily, and access data from anywhere.

CHATGPT

Blockchain technology is a digital system used to securely store and share information across multiple computers. Data is stored in blocks that are linked together in a chain, making it difficult to alter or hack. Blockchain is decentralized, meaning no single person or organization controls it, which improves security and transparency. It is widely used in cryptocurrencies like Bitcoin, but also has applications in banking, healthcare, supply chains, and voting systems. Blockchain helps create secure, reliable, and transparent digital transactions without the need for a central authority.

GEMINI

The Basics of Blockchain Technology

At its core, a blockchain is a shared digital record book that tracks information across a network of many computers. Unlike a traditional database owned by a single bank or company, blockchain is decentralized, meaning no single person or organization is in charge.

How It Works: Information is bundled into "blocks." Once a block is full of data, it is digitally "chained" to the previous block. Every computer in the network (called a node) holds a copy of the entire record. Before any new information is added, the computers in the network must agree that the data is valid (Consensus).

Why It Matters: High Security (changing old information is nearly impossible), Transparency, and Efficiency. The Bottom Line: Blockchain is a secure, transparent way to store data that ensures everyone sees the same "truth" without needing to rely on a central authority.

CLAUDE

Blockchain Technology - Quick Summary

Blockchain is a digital record-keeping system that stores information across many computers instead of one central location. Data is saved in "blocks" that are linked together in a "chain," making it nearly impossible to alter past records. It is secure, transparent, and does not need a bank or authority to verify transactions. Blockchain is widely used in digital currencies like Bitcoin, as well as in healthcare, supply chains, and legal contracts.

COPILOT

Blockchain Made Simple

What it is: Blockchain is like a digital notebook that everyone shares. Once something is written in it, it can't be erased or changed without everyone agreeing.

How it works: Information is grouped into "blocks." Each block is linked to the one before it, forming a secure "chain."

No single boss: Instead of being controlled by one company or government, blockchain is managed by many computers working together (decentralization).

Why it's safe: Special codes protect the data. If someone tries to change it, the whole network will notice.

Key takeaway: Blockchain is a powerful way to store and share information safely without needing a central authority. It's already changing industries, but it still faces challenges before becoming mainstream.

3. CHAIN OF THOUGHT

Prompt:

You are part of a content curation team for an educational platform that delivers quick summaries of research papers and technical articles to undergraduate students. Your task is to summarize a 500-word technical article on "The Basics of Blockchain Technology" in a clear, concise, and student-friendly manner.
Follow these steps before writing the final summary:
1. Identify the main topic of the article.
2. List the key concepts discussed in the article.
3. Determine the purpose and benefits of blockchain technology mentioned in the article.
4. Remove unnecessary technical details and complex jargon.
5. Rewrite the important points in simple language suitable for undergraduate students.
6. Combine the points into a short and coherent summary.

Finally, provide: A concise summary in 120-150 words, Simple and easy-to-understand language, Only the most important information from the article.

CHATGPT

Blockchain technology is a secure digital system used to store and share information across multiple computers. It works by recording data in blocks that are connected in a chain, making the information difficult to change or hack. One of the main features of blockchain is decentralization, which means no single person or organization controls the system. This improves security, transparency, and reliability.

Blockchain is mainly known for supporting cryptocurrencies like Bitcoin, but it is also used in banking, healthcare, supply chain management, and online voting systems. The technology helps make transactions safer, faster, and more transparent without needing a central authority. Overall, blockchain provides an efficient and trustworthy way to manage digital information and transactions in many industries.

GEMINI

Blockchain is a decentralized digital ledger that records transactions across a massive network of computers. Unlike a traditional bank that stores data in one central location, blockchain distributes its records globally, ensuring that no single person or entity has total control.

Information is grouped into "blocks," which are digitally "chained" together in a specific chronological order. Every computer in the network must agree on new data through a consensus process before it is added, making the system incredibly secure.

Key Benefits for Students to Know: Immutability (cannot be changed or deleted), Transparency (all participants can view the history), Efficiency (removing middlemen). Ultimately, blockchain provides a secure, "single source of truth" for everything from digital currency to supply chain tracking.

CLAUDE

Final Summary (120-150 words)

Blockchain is a shared digital record-keeping system that stores information across thousands of computers worldwide. Data is organized into "blocks," each linked to the previous one, forming a secure "chain." Once information is recorded, it cannot be changed or deleted, making the system highly reliable and tamper-proof.

Unlike traditional systems, blockchain has no central authority no bank, government, or company controls it. Instead, all participants in the network must agree before any new data is added, ensuring fairness and transparency.

Originally designed for digital currencies like Bitcoin, blockchain is now used in many fields including healthcare, supply chain management, voting systems, and legal contracts. Its biggest advantage is that it allows people to trust shared information without relying on a middleman. In short, blockchain makes digital interactions more secure, transparent, and independent.

COPILOT

Summary (120-150 words)

Blockchain is a digital system for recording information securely and transparently. Instead of being controlled by one authority, it works through a network of computers, making it decentralized. Data is stored in "blocks," and each block is linked to the previous one, forming a chain that is very hard to tamper with. Special codes protect the data, and everyone in the network can view the records, which builds trust.

Blockchain is best known for powering cryptocurrencies like Bitcoin, but it also supports smart contracts, supply chain tracking, and secure voting systems. Its main benefits include strong security, reduced need for middlemen, and improved efficiency. However, challenges remain, such as high energy use, slower performance for large systems, and unclear regulations. Key takeaway: Blockchain is a powerful tool for secure, transparent information sharing, with wide applications but ongoing limitations.

4. ROLE BASED PROMPT

Prompt:
You are an educational content curator working for an online learning platform that provides simplified research paper and technical article summaries for undergraduate students. Your role is to read and summarize a 500-word technical article on "The Basics of Blockchain Technology" in a way that is easy for students to understand.
While summarizing, you should:
- Identify the main ideas and important concepts
- Explain the content in simple, non-technical language
- Keep the summary concise and engaging
- Focus on the purpose, working, benefits, and applications of blockchain technology
- Avoid unnecessary details and complex jargon

Your final output should be: Clear and student-friendly, Around 120-150 words, Informative yet easy to read for undergraduate learners.

CHATGPT

Blockchain technology is a digital system that securely stores and shares information across many computers. It records data in blocks that are linked together, forming a chain. Once information is added, it cannot be easily changed, which makes blockchain secure and trustworthy. Unlike traditional systems,

blockchain is decentralized, meaning no single person or organization controls it. This improves transparency and reduces the risk of fraud or data loss.

Blockchain is mainly used in cryptocurrencies like Bitcoin, but it also has applications in banking, healthcare, supply chain management, and online voting. The technology helps make transactions faster, safer, and more reliable without requiring a central authority. Overall, blockchain is an innovative technology that improves the security and efficiency of storing and exchanging digital information.

GEMINI

Blockchain Basics: A Simple Guide

Blockchain is essentially a shared digital record book that tracks information across a network of computers. Unlike a traditional bank database that is controlled by one institution, blockchain is decentralized, meaning no single person or company owns it.

How It Works: Data is stored in "blocks." Once a block is full, it is linked to the previous one, creating a chronological "chain." Before any new information is added, the computers in the network must agree it is valid. This process makes the records nearly impossible to fake or delete.

Why It Matters: Trust (no need for a "middleman"), Security (changing info requires hacking thousands of computers at once), Applications (Beyond Bitcoin, it is used for secure voting, tracking shipping goods, and protecting medical records).

CLAUDE

Final Student-Friendly Summary

Blockchain is a digital system that securely records and stores information across a large network of computers. Think of it as a shared notebook that thousands of people hold simultaneously once something is written, nobody can erase or change it.

Information is stored in "blocks" that are connected in a "chain," keeping records organized, transparent, and tamper-proof. No single person, bank, or government controls it - making it fair and open for everyone involved. Before any new information is added, the majority of participants must approve it, which keeps the system honest and trustworthy.

While blockchain powers popular cryptocurrencies like Bitcoin, it has grown far beyond digital money. Today, it is transforming healthcare, supply chain tracking, online voting, and legal agreements by making data sharing safer and eliminating the need for middlemen. Blockchain, simply put, is the technology that makes digital trust possible.

COPILOT

Summary (120-150 words)

Blockchain is a digital system that records information in a secure and transparent way. Instead of being controlled by one central authority, it operates through a network of computers, making it decentralized. Data is stored in "blocks," and each block is linked to the one before it, forming a chain that is very difficult to change. This structure, combined with cryptography, ensures strong security and trust because everyone in the network can see and verify the records.

Blockchain is widely known for supporting cryptocurrencies like Bitcoin, but its uses go beyond digital money. It enables smart contracts that run automatically, helps track goods in supply chains, and can make voting systems more reliable. The main benefits include security, efficiency, and reduced need for middlemen. However, challenges such as high energy use, slower performance, and unclear regulations remain. Key takeaway: Blockchain is a powerful tool for secure, transparent information sharing with growing real-world applications.

5. EVALUATION METRICS & CONCLUSION

Prompt Technique	AI Platform	Accuracy (1-5)	Coherence (1-5)	Simplicity (1-5)	Speed (1-5)	UX (1-5)
Zero-shot Prompting	ChatGPT	4	4	5	5	4
	Gemini	5	4	3	4	4
	Claude	5	5	4	3	5
	Copilot	4	4	4	4	5
Few-shot Prompting	ChatGPT	4	4	5	5	4
	Gemini	5	5	4	3	5
	Claude	5	5	5	3	5
	Copilot	4	4	4	4	5
Chain-of-Thought Prompting	ChatGPT	4	5	4	4	4
	Gemini	5	5	4	3	5
	Claude	5	5	5	4	5
	Copilot	4	4	4	4	5
Role-based Prompting	ChatGPT	4	5	5	5	5
	Gemini	5	5	5	4	5
	Claude	5	5	5	4	5
	Copilot	4	4	5	4	5

Final Verdict

Best Prompt Technique: Chain-of-Thought Prompting
Based on the analysis, Chain-of-Thought prompting performed the best.

Best AI Model: Claude
Claude performed the best overall.

Best Combination: Claude + Chain-of-Thought Prompting
This combination gave the strongest results overall.